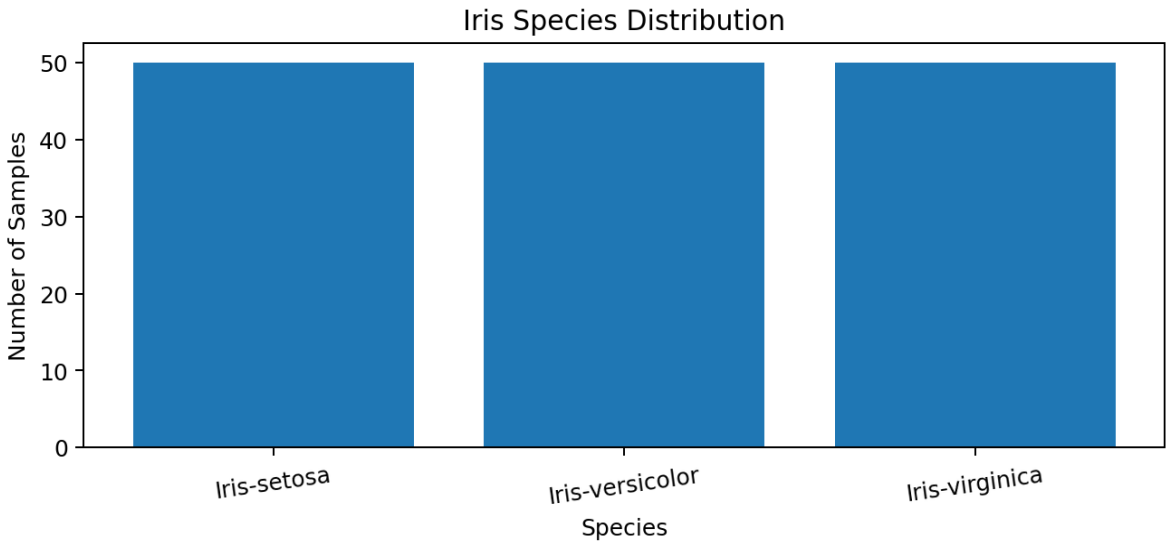
IRIS FLOWER CLASSIFICATION

Machine Learning Internship — Task 1

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Objective: Predict an Iris flower species using sepal and petal measurements with supervised machine learning.

Project item	Details
Dataset	150 samples, 4 numerical features + species target
Algorithms	k-Nearest Neighbors (k-NN), Logistic Regression, Decision Tree
Best reported model	k-Nearest Neighbors (k-NN)
Best reported accuracy	100.00% on the selected test split

GitHub repository and Google Colab notebook links are included in the submission section.

1. Executive Summary

This project develops a supervised machine-learning classification workflow to predict the species of an Iris flower from four measurements: sepal length, sepal width, petal length, and petal width. The workflow includes dataset loading, exploratory data analysis (EDA), target encoding, train-test splitting, model training, evaluation, comparison, and saving a reusable model.

The project documentation reports 150 samples and 5 columns, with 50 samples per species. Three algorithms were compared. The recorded accuracies are 100.00% for k-NN, 96.67% for Logistic Regression, and 93.33% for Decision Tree. k-NN was selected as the best-performing model on the selected test split and saved as **iris_knn_model.joblib**.

Key result: the documented sample input [5.1, 3.5, 1.4, 0.2] was predicted as Iris-setosa.

2. Project Objectives

• Understand and inspect the Iris dataset.
• Check missing values and review feature statistics.
• Visualize species distribution and relationships among measurements.
• Train and compare three classification algorithms.
• Evaluate model performance using accuracy and classification metrics.
• Save the selected model and demonstrate prediction on a new flower sample.

Tools and Libraries

Tool / Library	Purpose
Python	Main programming language
Pandas and NumPy	Data manipulation and numerical operations
Matplotlib and Seaborn	Charts and exploratory data analysis
Scikit-learn	Data splitting, classification models, and evaluation
Joblib	Saving and loading the trained model
Google Colab	Cloud notebook environment used for development

3. Dataset Information

The Iris dataset contains 150 flower observations and five columns. Four numerical measurements are used as model inputs; the species column is the target to be predicted.

Dataset item	Description
Number of records	150
Number of columns	5
Input features	sepal_length, sepal_width, petal_length, petal_width
Target variable	species
Classes	Iris-setosa, Iris-versicolor, Iris-virginica
Class balance	50 samples for each species
Missing values	The project notebook reports no missing values

4. Exploratory Data Analysis (EDA)

EDA was performed to understand data quality, feature ranges, class balance, and the relationship between flower measurements. Visualizations included species counts, scatter plots, pair plots, and box plots.

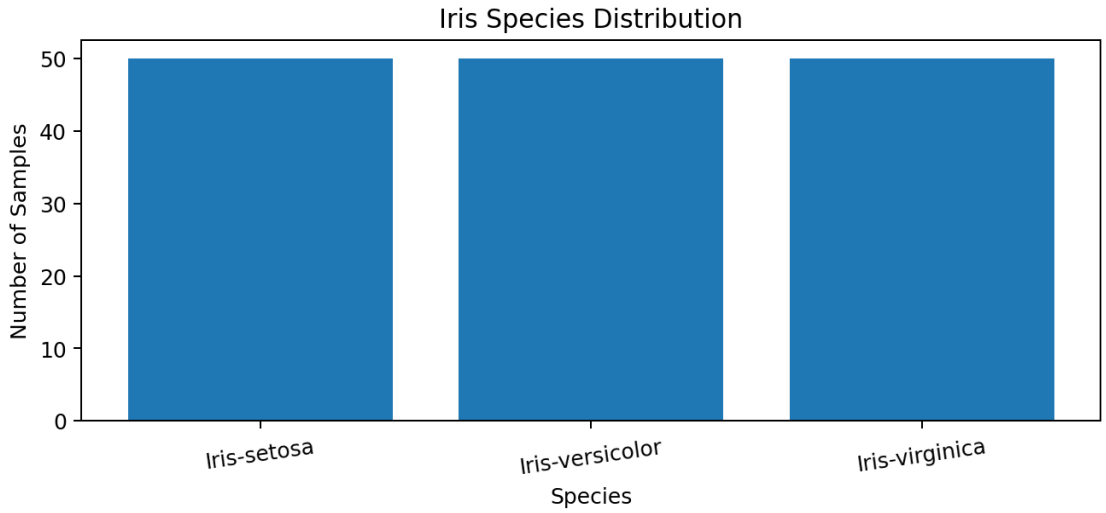


Figure 1. Species distribution. The classes are balanced with 50 samples per species.

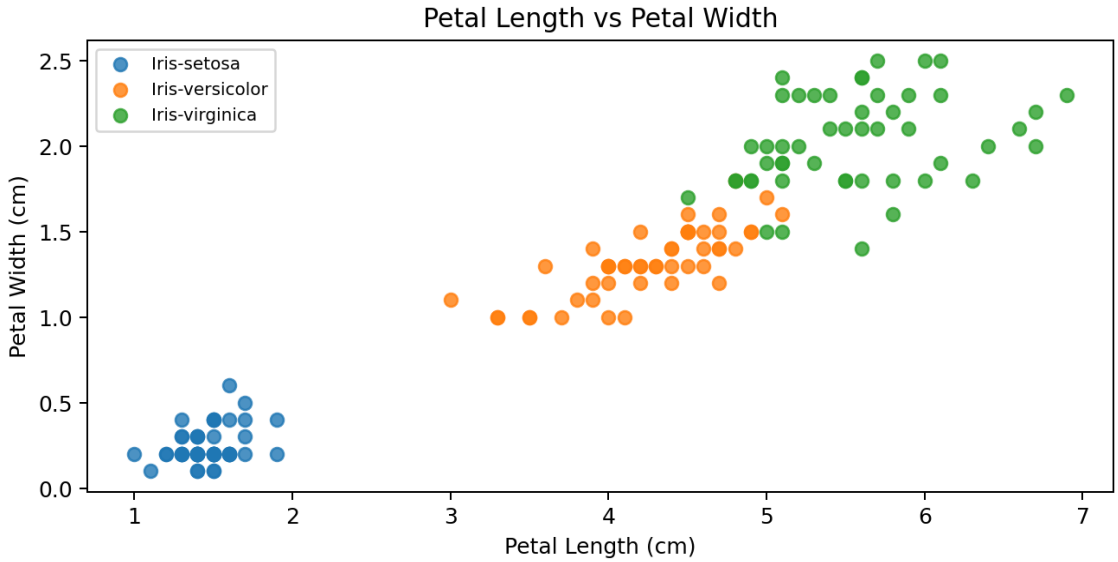


Figure 2. Petal length versus petal width. The plotted measurements show visible separation between Iris-setosa and the other species.

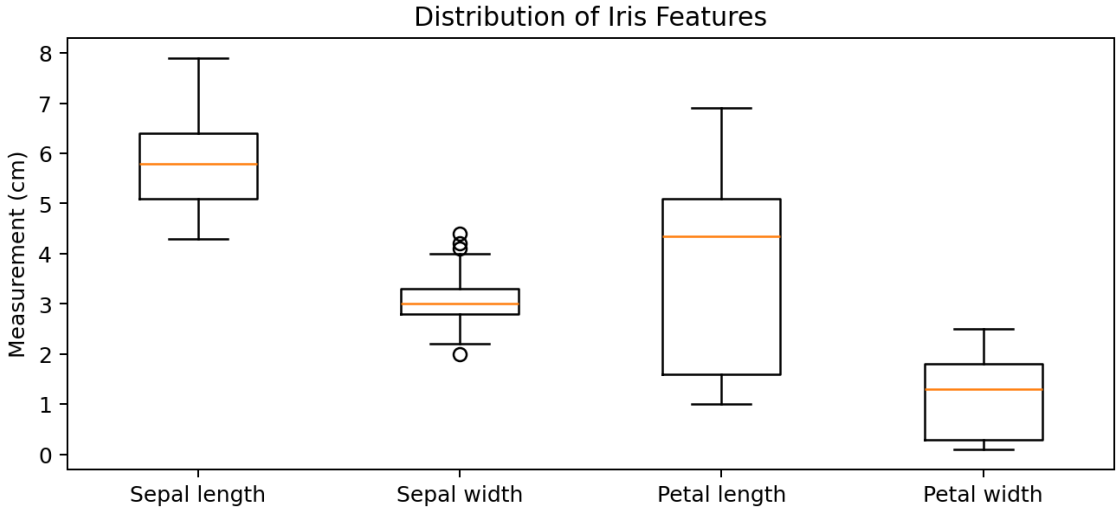


Figure 3. Box plots summarizing the distribution and spread of the four numerical features.

5. Methodology and Data Preparation

The project follows a supervised machine-learning workflow from data inspection to model inference.

Step	Activity
1. Load dataset	Read the Iris data into a Pandas DataFrame.
2. Inspect data	Review dimensions, data types, missing values, and summary statistics.
3. Explore and visualize	Plot class distribution and feature relationships.
4. Encode target	Map the three species to numeric labels for model training.
5. Select features	Use sepal length, sepal width, petal length, and petal width as inputs.
6. Split data	Use an 80:20 training/testing split as documented in the project.
7. Train classifiers	Fit k-NN, Logistic Regression, and Decision Tree models.
8. Evaluate models	Compare accuracy and review classification metrics.
9. Save model	Save the selected k-NN model as iris_knn_model.joblib.
10. Predict	Load/use the model to predict the species of an example flower.

Target label encoding

Species	Encoded label
Iris-setosa	0
Iris-versicolor	1
Iris-virginica	2

The labels are numeric representations for model training; the final prediction is interpreted back as the species name.

6. Model Training and Comparison

Three classification algorithms were trained and compared using the performance results recorded in the project documentation.

Model	Configuration / description	Reported accuracy
k-Nearest Neighbors (k-NN)	n_neighbors = 5	100.00%
Logistic Regression	max_iter = 200	96.67%
Decision Tree	random_state = 42	93.33%

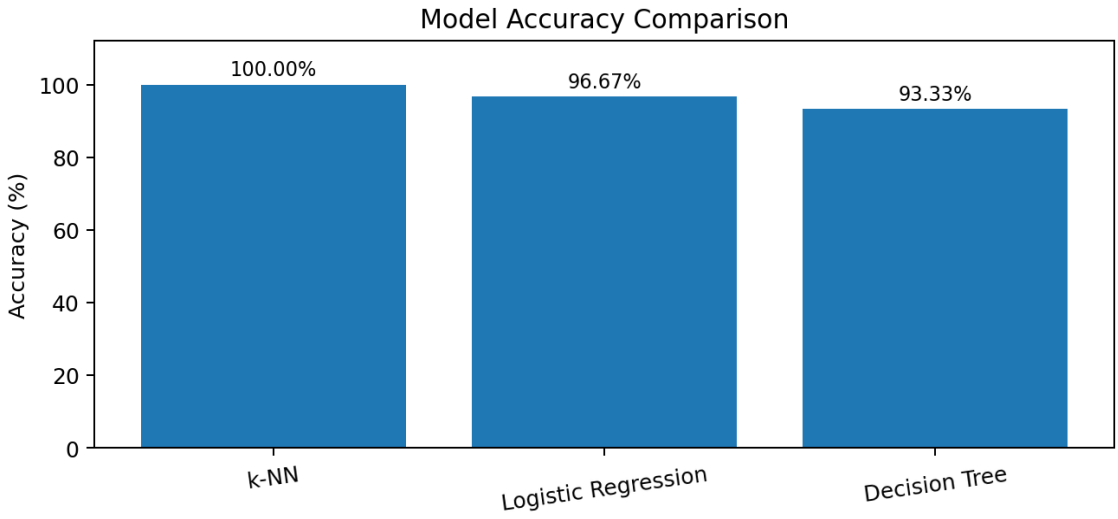


Figure 4. Accuracy comparison based on the results recorded in the project notebook/report.

Model selection

k-NN achieved the highest recorded test accuracy and was selected as the final model. This result applies to the particular train-test split used in the project; it should not be interpreted as a guarantee of 100% accuracy on every future dataset.

7. Model Evaluation

Accuracy: the proportion of test examples predicted correctly.
Precision: how often a predicted class is correct.
Recall: how many examples of an actual class are identified correctly.
F1-score: a combined measure of precision and recall.
Confusion matrix: a table showing correct and incorrect predictions for each class.

Model	Precision (macro avg.)	Recall (macro avg.)	F1-score (macro avg.)	Accuracy
k-NN	1.00	1.00	1.00	1.0000
Logistic Regression	0.97	0.97	0.97	0.9667
Decision Tree	0.93	0.93	0.93	0.9333

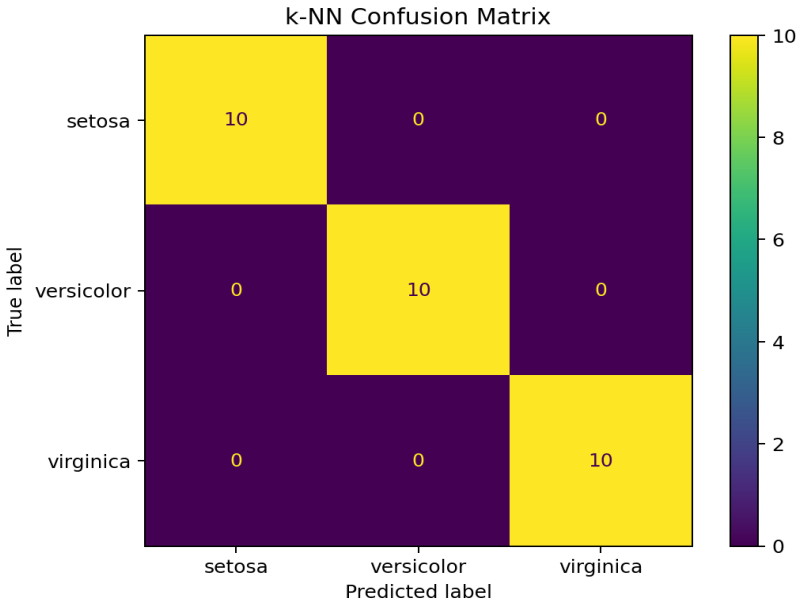


Figure 5. Recreated k-NN confusion matrix using an Iris dataset 80:20 stratified split with random_state=42. The accuracy values in the comparison table follow the documented project results.

8. Sample Prediction and Saved Model

The project saves the selected k-NN classifier with Joblib so it can be loaded and reused for predictions.

Input feature	Sample value
Sepal length	5.1 cm
Sepal width	3.5 cm
Petal length	1.4 cm
Petal width	0.2 cm

Recorded sample prediction: Iris-setosa

Saved artifact

File	Purpose
iris_knn_model.joblib	Saved k-NN classifier for later inference
requirements.txt	Lists the Python packages required by the project
Iris_Flower_Classification_Task_1.ipynb	Notebook containing code and outputs
README.md	Project description and usage notes

The saved model and notebook should be kept together with the report and requirements file for reproducibility and internship submission.

9. Project Links and Submission Deliverables

Google Colab Notebook:
https://colab.research.google.com/drive/1bHOJGv5ueIx4u3kaxBIVe7MsGdT4cqNR?usp=sharing

GitHub Repository:
https://github.com/RushikeshGame/Iris-Flower-Classification

Deliverables checklist

Deliverable	Purpose / status
Google Colab notebook	Code, execution cells, and outputs
GitHub repository	Project source and documentation
iris_knn_model.joblib	Saved trained model
requirements.txt	Python dependency list
Final report PDF	Project overview, methodology, visuals, evaluation, prediction, and links

10. Conclusion

The Iris Flower Classification project demonstrates a complete beginner-level supervised machine-learning workflow, including dataset inspection, EDA, visualization, label encoding, train-test splitting, model training, evaluation, comparison, saving, and inference. The documented results show k-NN with the highest accuracy at 100.00%, followed by Logistic Regression at 96.67% and Decision Tree at 93.33%. The selected k-NN model was saved as **iris_knn_model.joblib**, and the sample input [5.1, 3.5, 1.4, 0.2] was recorded as Iris-setosa.

Validation note: reported accuracy describes the selected test split. Additional cross-validation or independent test data would provide a stronger estimate of performance on unseen examples.

END OF REPORT